

Measures of Centre and Spread

Data Analysis

2026

Preliminary Mathematics Standard

Outline

Measures of Centre

Shape of Distributions

Measures of Spread

Comparing Datasets

Choosing the Right Measure

Practice Questions

Homework

Table of Contents

Measures of Centre

Shape of Distributions

Measures of Spread

Comparing Datasets

Choosing the Right Measure

Practice Questions

Homework

What is a Measure of Centre?

A **measure of centre** describes a typical or central value in a dataset.

There are three main measures:

- **Mean** — the arithmetic average
- **Median** — the middle value when data is ordered
- **Mode** — the most frequently occurring value

Definition: The sum of all values divided by the number of values.

$$\bar{x} = \frac{\sum x}{n}$$

- Uses every value in the dataset
- Affected by outliers

Example

Dataset: 4, 7, 3, 9, 2, 7, 6

$$\bar{x} = \frac{4 + 7 + 3 + 9 + 2 + 7 + 6}{7} = \frac{38}{7} \approx 5.43$$

Median

Definition: The middle value of an ordered dataset.

+ not affected by outliers

Finding the median:

- Order data from smallest to largest
- If n is **odd**: median is the $\frac{n+1}{2}$ th value
- If n is **even**: median is the average of the $\frac{n}{2}$ th and $\left(\frac{n}{2} + 1\right)$ th values

Example

Ordered dataset: 2, 3, 4, 6, 7, 7, 9 ($n = 7$, odd)

$$\text{Median} = 4\text{th value} = 6$$

Definition: The value that appears most often in a dataset.

*multimodal
↳ 6: mode)*

- A dataset can have **no mode**, **one mode**, or **multiple modes**
- The only measure of centre usable with **categorical data**
- Not affected by outliers

Example

Dataset: 2, 3, 4, 6, 7, 7, 9

Mode = 7 (appears twice)

Calculating from a Frequency Table

Value (x)	Freq (f)
2	3
4	5
6	2
8	4
Total	14

Example

$$\begin{aligned}\bar{x} &= \frac{\sum fx}{\sum f} = \frac{2(3) + 4(5) + 6(2) + 8(4)}{14} \\ &= \frac{70}{14} = 5\end{aligned}$$

Mode = 4 (highest frequency)

Median: average of 7th and 8th values — both in the “4” group
 $\therefore$ Median = 4

Table of Contents

Measures of Centre

Shape of Distributions

Measures of Spread

Comparing Datasets

Choosing the Right Measure

Practice Questions

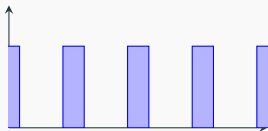
Homework

Datasets are described by the number of peaks they have:

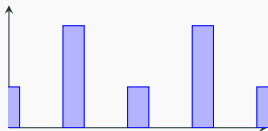
- **Uniform** — all values occur with roughly equal frequency; no clear peak
- **Unimodal** — one clear peak
- **Bimodal** — two peaks of roughly equal height
- **Multimodal** — more than two peaks

Distribution Shapes — Diagrams

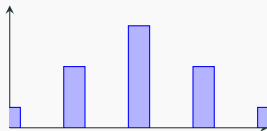
Uniform



Bimodal



Unimodal



Multimodal

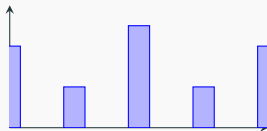


Table of Contents

Measures of Centre

Shape of Distributions

Measures of Spread

Comparing Datasets

Choosing the Right Measure

Practice Questions

Homework

What is a Measure of Spread?

A **measure of spread** describes how varied or dispersed the values in a dataset are.

Two key measures:

- **Range** — simplest measure; difference between maximum and minimum
- **Standard deviation** — how far values typically deviate from the mean

A **small** spread $\Rightarrow$ values are close together.

A **large** spread $\Rightarrow$ values are more variable.

Definition:

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

- Simple to calculate
- Only uses two values — heavily affected by outliers
- Does not describe how the middle data is distributed

Example

Dataset: 2, 3, 4, 6, 7, 7, 9

$$\text{Range} = 9 - 2 = 7$$

Population Standard Deviation

Definition

The average distance of each value from the mean.

$$\text{S.D.} \rightarrow \sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}}$$

Example

Dataset: 3, 5, 7, 7, 8

($\bar{x} = 6$)

$$\frac{3 + 5 + 7 + 7 + 8}{5} = 6$$

mean

x	$x - \bar{x}$	$(x - \bar{x})^2$
3	-3	9
5	-1	1
7	+1	1
7	+1	1
8	+2	4
		$\Sigma = 16$

$$\sigma = \sqrt{\frac{16}{5}} = \sqrt{3.2} \approx 1.79$$

Table of Contents

Measures of Centre

Shape of Distributions

Measures of Spread

Comparing Datasets

Choosing the Right Measure

Practice Questions

Homework

Comparing Datasets

To compare two datasets, examine both their **centre** and **spread**.

Example

	Dataset A	Dataset B
Values	4, 5, 6, 6, 7, 8	1, 3, 6, 8, 10, 12
Mean	6.00	6.67
Median	6.00	7.00
Range	4	11
σ	1.29	3.86

Both datasets have similar centres, but Dataset B has a much larger spread — its values are far more variable.

Table of Contents

Measures of Centre

Shape of Distributions

Measures of Spread

Comparing Datasets

Choosing the Right Measure

Practice Questions

Homework

Merits of Each Measure of Centre

Measure	Use when...	Avoid when...
Mean	Data is symmetric with no outliers	Outliers or skewed data present
Median	Outliers or skewed data present	A measure using all values is needed
Mode	Categorical data; finding the most common value	Data has no repeated values

Real-World Examples

Appropriate uses:

- **Median** for house prices — a few very expensive homes skew the mean
- **Mode** for shoe sizes — most common size determines stock orders
- **Mean** for exam scores in a balanced class — gives a fair overall picture

Inappropriate uses:

- **Mean** for income data — billionaires distort the average upward
- **Mode** for continuous data (e.g. heights) — values rarely repeat exactly
- **Range** alone for spread — one extreme value makes it misleading

Table of Contents

Measures of Centre

Shape of Distributions

Measures of Spread

Comparing Datasets

Choosing the Right Measure

Practice Questions

Homework

Practice — Measures of Centre

Example

The following data shows the number of hours 8 students studied last week:

3, 5, 5, 7, 8, 9, 9, 12

1. Calculate the **mean**.
2. Find the **median**.
3. State the **mode**.
4. Describe the **shape** of this distribution.

bimodal, symmetric

Practice — Frequency Table

Example

Score	Frequency
10	2
20	6
30	4
40	3
50	1

Using the frequency table:

1. Calculate the **mean score**.
2. Find the **median score**.
3. State the **mode**.

Practice — Measures of Spread

Example

Two athletes record their sprint times (seconds) over 6 trials:

Athlete A: 11.2, 11.4, 11.3, 11.5, 11.2, 11.4

Athlete B: 10.8, 12.1, 11.0, 11.9, 10.5, 12.7

$$\sigma = \sqrt{\frac{\sum (x - \bar{x})^2}{n}} > 0$$

1. Calculate the **mean** for each athlete.
2. Calculate the **range** for each athlete.
3. Calculate the **population standard deviation** for each athlete.
4. Which athlete is more **consistent**? Justify your answer.

$$\sigma_A < \sigma_B$$

Practice — Choosing the Right Measure

Example

1. A real estate agent reports the mean house price in a suburb as \$1 250 000. Most houses sell for around \$800 000, but one recently sold for \$6 500 000.
~~~~~ ← outlier
 - (a) Explain why the mean is misleading here.
 - (b) Which measure of centre would be more appropriate? Why?
Median
2. A café sells 5 sizes of coffee. Last week's sales were:
Small (12), Medium (45), Large (38), Extra Large (20), Jumbo (5).
 - (a) Which measure of centre is most appropriate? *Mode*
 - (b) State its value.
Medium

Table of Contents

Measures of Centre

Shape of Distributions

Measures of Spread

Comparing Datasets

Choosing the Right Measure

Practice Questions

Homework

Homework

Complete all questions from *Data Analysis — Measures of Centre and Spread*.